

HealthConnect Project Summary

Date: 11/09/2026

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Dataset: HealthConnect Appointment Dataset

What I planned to accomplish

- Dig deeper into the analysis
- Test hypothesis
- Refine dashboards to accommodate change
- Present insights to other tracks and note any improvements

What I completed

- Completed deeper analysis by statistically validating the key drivers of no-shows and KPI's
- I introduced and validated the new "Previous No-show Rate" KPI.
- Provided a list of high priority, actionable recommendations.
- Created a new dashboard for deeper analysis & risk assessment
- Shared my insights with the data science & project management track

What I improved from Week 5

- The analysis has evolved from descriptive EDA to statistically validated prescriptive decision support. The findings are now backed by hypothesis tests and confidence intervals, making them far more reliable
- I also created lead_days_group & previous appointment group fields for easier analysis
- Removed previous no show KPI

What I integrated

I performed a cross-track integration with the Data Science track, providing them with validated features and a new KPI to enhance their predictive model.

Which track(s) I collaborated with

Data Science

What was exchanged

I provided validated findings and in return, I will receive model feedback on feature importance.

What changed as a result

My analytical findings are no longer just observations; they are now specific, prioritized inputs for a predictive model, directly guiding the Data Science track's feature engineering and model refinement. This creates a direct, evidence-based link between our two tracks.

Key findings or development outcomes

- The no-show problem is driven by a combination of logistical barriers, patient history, and demographic factors.
- The "Previous No-show Rate" is a superior risk metric.
- Reminders alone are insufficient for overcoming the strongest barriers.

Major challenges

Ruling out confounding variables to establish the true driver of no-shows for the distant patient group.

Important decisions

- To adopt "Previous No-show Rate" as a new key KPI
- revise the Week 5 hypothesis about waiting time being a predictive factor.

Remaining issues

The dataset's snapshot nature limits our ability to measure the impact of any new interventions.

My contribution to the overall HealthConnect project

My contribution has shifted from an initial EDA to a validated decision support report. The advanced analytics and refined KPIs will serve as the foundation for a more accurate no-show prediction model and more targeted business interventions, directly addressing the project's core question.

Proposed focus for Week 7

To support the Data Science track in testing their model by providing further analytical validation of feature importance.

I will also work on updating the dashboard to include the new KPI and visualizations identified this week, ensuring it is fully aligned with our validated findings.